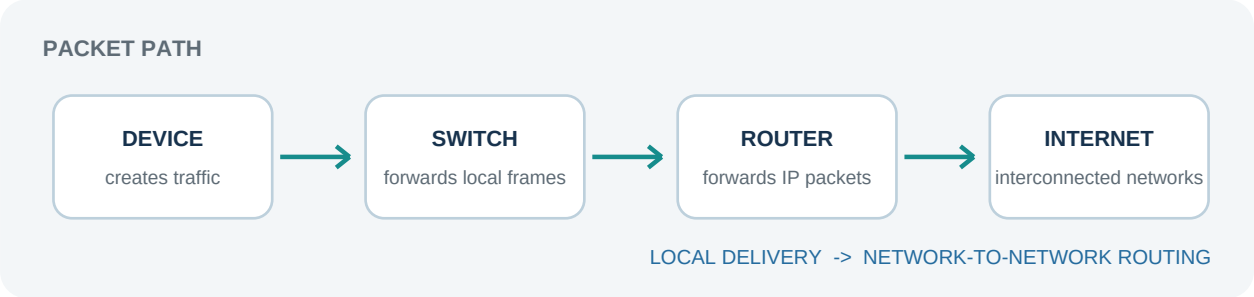
NETWORKING STUDY GUIDE

Foundations, switching, routing and address resolution



LEARNING IN PROGRESS | Notes cover the first three recorded NetworkChuck videos and Lecture 01 of the primary lecture series. This guide is a revision aid, not a claim of course completion or mastery.

Primary source
CS-340 Intro to Computer Networking (Steve Tarzia) - Lecture 01: Introduction. This course is the main learning path and will guide deeper future additions as study progresses. The series is reported as 18 lectures; Lecture 02 is in progress.

Supporting source
NetworkChuck FREE CCNA 200-301 playlist - the first three videos are recorded as completed for these notes. Remaining playlist material is upcoming.

A practical reference for understanding how devices connect, how local switches forward frames, and how routers move packets between networks.

Contents

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NO.	TOPIC	WHAT YOU WILL FIND
00	Learning scope	Primary course, supporting videos and recorded progress
01	Network fundamentals	Internet, access, protocols, switching models and performance
02	Switching basics	Frames, MAC learning and read-only inspection
03	Routing and ARP	Gateways, local versus remote traffic and route inspection

Core distinction
Switches forward local Ethernet frames. Routers forward IP packets between networks.

Learning scope and source map

CS-340 Intro to Computer Networking, taught by Steve Tarzia, is the primary learning source for this section. Lecture 01 anchors the foundations chapter; future lectures will be added only after their material has been studied. NetworkChuck videos supplement the early practical introduction to networks, switching and routing.

Source / status	Coverage recorded here
Primary: CS-340 Intro to Computer Networking (Steve Tarzia)	Lecture 01 - Introduction: Internet overview, protocols, access networks, network core, switching models and introductory performance measures. Lecture 02 is in progress; Lectures 03-18 are upcoming.
Supporting: NetworkChuck FREE CCNA	Videos 1-3 recorded completed: network basics, switches, routers/ARP. Remaining videos are not documented as studied.
Boundary	The supplied OSI/TCP-IP video is not yet covered in this notes set. Optional labs and measurements are not marked complete.

What each source contributes

- Lecture 01 gives the broader conceptual frame: edge and core, access networks, protocol rules, packet versus circuit switching, and performance vocabulary.
- NetworkChuck offers a beginner-friendly bridge into local switching and routing examples, including MAC learning, ARP, and default-gateway behavior.
- The guide combines the verified scope into one mental model while keeping the learning-status record explicit.

Primary source: CS-340 Intro to Computer Networking - Lecture 01 (Steve Tarzia) | CS-340 playlist - 18 videos

Supporting source: NetworkChuck FREE CCNA playlist

1. Internet and network fundamentals

A computer network is a set of connected devices and links that exchange information using shared rules called protocols. The Internet is a global interconnection of independently operated networks. The Web is one service on the Internet; email and voice services are others.

The end-to-end path

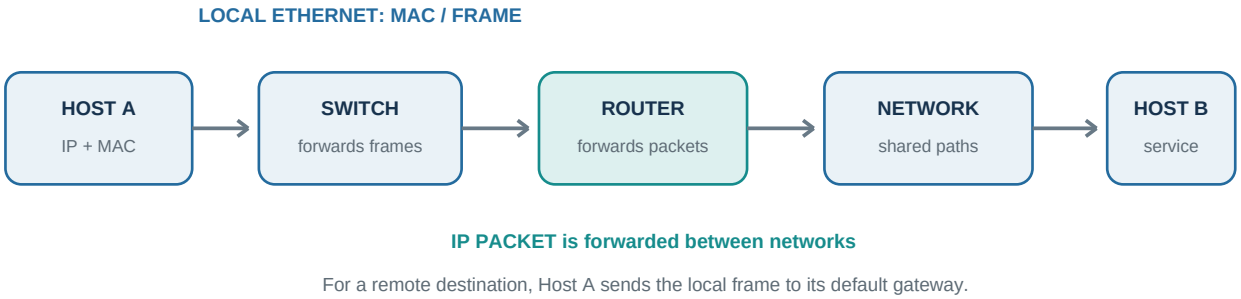

Figure 1. A simplified path. Real traffic can cross more networks and routers; routes can change. A switch forwards local Ethernet frames, while a router forwards IP packets between networks.

Useful terms

Term	Meaning
Network edge	End devices and services that create or receive traffic.
Access network	The connection from a user or organization to a provider network.
LAN	A local network serving a home, office or campus area.
Network core	Interconnected provider/backbone networks that carry traffic between edge networks.
ISP	A provider that supplies network connectivity and may connect customers to other networks.
Protocol	Rules for message format, meaning and expected actions between communicating systems.

Access media and examples

- Guided media carry signals over a physical path such as copper or fiber. Unguided media use radio, as Wi-Fi does.
- The primary lecture outline introduces PSTN, DSL and cable access as examples of how Internet access has evolved. DSL uses telephone-line infrastructure; cable access commonly uses television coaxial infrastructure.
- Frequency-division multiplexing (FDM) divides available spectrum into frequency bands so multiple signals can share a medium.

Circuit switching and packet switching

	Circuit switching	Packet switching
How it works	Establishes a path with resources reserved for a session.	Splits data into packets that share links with other traffic.
Resource use	Capacity can remain reserved during the session.	Links are shared as packets are forwarded.
Context	Traditional telephone-network designs.	General Internet data transport model.

Packet paths and delays may vary. Shared links use capacity efficiently, but congestion can affect experienced performance. Real systems add mechanisms for reliability, service quality and traffic engineering.

Performance vocabulary

Measure	What it describes
Bandwidth / link capacity	Maximum data rate in a stated context.
Throughput	Data rate actually achieved by a flow or application.
Latency / delay	Time for data to travel; round-trip time includes the return path.
Jitter	Variation in packet delay over time.
Packet loss	Packets that fail to reach the intended receiver.
CDF	A cumulative distribution function summarizes the fraction of observations at or below each measured value.

Example: A video call may have sufficient average throughput yet feel poor when delay varies or packets are lost. Repeat measurements and record where and when they were taken; one measurement does not describe a changing network.

Lecture timestamps

The published lecture outline links directly to: Internet overview, 4:45 | Internet vs Web, 21:48 | access networks and LANs, 33:01 | DSL and cable, 44:41 | network core, 56:24 | packet switching, 1:02:05 | performance, 1:06:24

Host-side observation

These commands request diagnostic information. Replace placeholders with a local gateway or a destination you are authorized to test. Output depends on the host and network.

```
Windows:  ipconfig
          ping -n 4 <your-default-gateway>
          tracert <authorized-destination>

Linux:    ip address
          ip route
          ping -c 4 <your-default-gateway>
          traceroute <authorized-destination>
```

2. Switching basics

A switch connects devices on a LAN and forwards Ethernet frames using destination MAC addresses. It learns which port can reach each source MAC address.

Basic forwarding behavior

Destination	Typical action
Known unicast MAC	Forward through the associated port within the relevant Layer 2 network.
Unknown unicast MAC	Flood to other eligible ports in that Layer 2 network.
Broadcast MAC	Flood to other eligible ports in that Layer 2 network.

Exact behavior depends on switch configuration, VLANs and platform. VLANs are outside the current notes scope.

Switch versus hub

Hub	Repeats incoming electrical signals to other ports; does not learn MAC addresses or selectively forward frames.
Switch	Examines Ethernet frames and uses learned MAC-to-port information to forward more selectively.

Switches forward local frames; routers forward IP packets between networks. Some devices combine both functions.

Example: two devices on one LAN

1. Device A sends a frame; the switch receives it on A's port.
2. The switch learns or refreshes the source MAC-to-port mapping.
3. It checks the destination MAC. If known, it forwards to the associated port; otherwise, it floods within eligible ports.
4. When device B replies, the switch can learn B's source MAC and port.

The switch makes this forwarding decision at the link layer.

Read-only inspection

```
Cisco IOS-style (device syntax varies):
  show mac address-table
```

This supplemental command displays MAC table entries; it is not claimed as the exact command used in the video. Use only a device or simulator you are authorized to access. See Cisco command reference.

Security application

Switching concepts help explain local traffic paths, device discovery and broadcast behavior. Unexpected MAC-to-port changes may merit investigation, but a MAC address can be changed or spoofed; a table entry alone does not establish a person's identity or malicious activity.

Remember: A switch MAC table does not route IP packets or identify a person. Broadcast and unknown destinations may be flooded.

3. Routing and address resolution

A router connects IP networks and forwards packets between them. It examines the destination IP address and consults a routing table to select an outgoing interface and, when needed, a next hop. A host's default gateway is the router it uses when a destination is not on the host's local IP network.

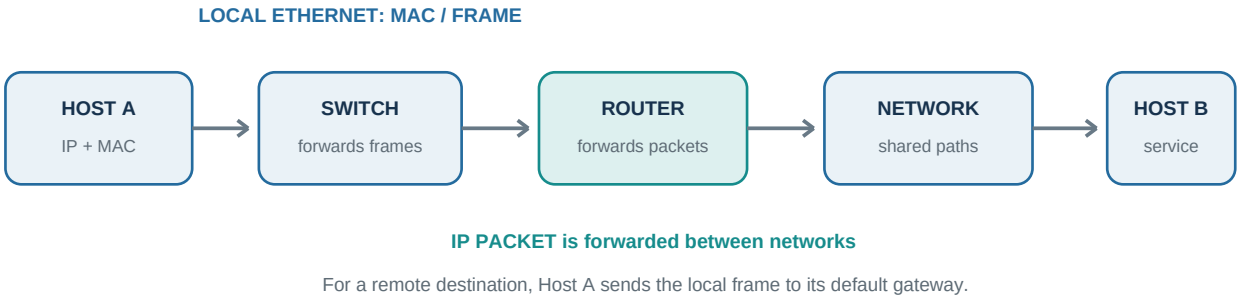

Figure 2. The switch handles local frame delivery; the router moves the IP packet toward another network.

Local or remote destination

- The host uses its IP address and network mask to decide whether a destination is local or remote.
- For a local destination, the host sends a frame directly to the destination's link-layer address.
- For a remote destination, it sends a local Ethernet frame to the default gateway's MAC address. The IP packet inside remains addressed to the final destination.

As a packet crosses a router, the router uses the IP destination and its routes to choose the next path. On an outgoing Ethernet link it builds a new frame for the next hop. Without address translation, the local frame addresses can change at each routed hop while the end-to-end IP destination remains the destination host.

ARP: local IPv4 address resolution

Address Resolution Protocol (ARP) lets an IPv4 device on an Ethernet local network find the link-layer address associated with a local IPv4 address. The device can broadcast an ARP request on its local link and cache the reply temporarily. See RFC 826 - ARP

For remote traffic, the host resolves the gateway's local IPv4 address; it does not ARP for the remote host's MAC address across the Internet. Routers do not forward ordinary Ethernet broadcasts between networks. IPv6 uses Neighbor Discovery rather than ARP.

Walkthrough: Host A reaches a host on another network

1. Host A determines that Host B is remote.
2. Host A consults its default gateway setting.
3. If needed, it uses ARP to learn the gateway MAC address on the local link.
4. It sends an Ethernet frame to the gateway MAC; the enclosed IP packet is addressed to Host B.
5. The router checks its route table and forwards toward the destination network.

These are linked but different decisions: the switch forwards the local frame based on MAC information; the router forwards the packet using IP routing information.

Safe read-only commands

```
Cisco IOS-style router:
  show ip interface brief
  show ip route
  show ip arp

Linux host:
  ip route
  ip neigh

Windows host:
  route print
  arp -a
```

These commands inspect interfaces, routes or cached neighbor entries. They do not change configuration. Syntax and output vary; run them only on systems you are authorized to inspect. Cisco reference: show ip route.

DNS preview: distinguish it from ARP

The third video briefly previews DNS. At a high level, DNS looks up records associated with names. DNS is not ARP: DNS is a naming system, while ARP resolves a local IPv4 address to a link-layer address. Detailed DNS study is not part of this guide.

Common mistakes and security context

- Assuming routers forward based only on a destination MAC; routing uses IP destinations and routes.
- Assuming ARP discovers a remote host's MAC through routers. ARP is local-link IPv4 resolution.
- Confusing the default gateway with a DNS resolver.
- Treating a route-table entry as proof that the full end-to-end path is healthy.

Revision check

When does a host use its gateway? Why does it ARP for that gateway? Which addresses identify the local frame and the final IP packet? How are ARP and DNS different?

Source video: What is a ROUTER? - FREE CCNA - EP 2 | ARP chapter, 4:16 | routers between networks, 9:44

To be continued...